

Copy-Paste to Mitigate Large Language Model Hallucinations

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Abstract—While Retrieval-Augmented Generation (RAG) enables large language models (LLMs) to generate contextually grounded responses, contextual faithfulness remains challenging as LLMs may not consistently trust provided context, leading to hallucinations that undermine reliability. We observe an inverse correlation between response copying degree and context-unfaithful hallucinations on RAGTruth, suggesting higher copying degrees reduce hallucinations by fostering genuine contextual belief. We propose CopyPasteLLM, obtained through two-stage high-copying response preference training. We design three prompting methods to enhance copying degree, demonstrating that high-copying responses achieve superior contextual faithfulness and hallucination control. These approaches enable a fully automated pipeline that transforms generated responses into high-copying preference data for training CopyPasteLLM. On FaithEval, ConFiQA and PubMedQA, CopyPasteLLM achieves best performance in both counterfactual and original contexts, remarkably with 12.2% to 24.5% accuracy improvements on FaithEval over the best baseline, while requiring only 365 training samples—1/50th of baseline data. To elucidate CopyPasteLLM’s effectiveness, we propose the *Context-Parameter Copying Capturing* algorithm. Interestingly, this reveals that CopyPasteLLM recalibrates reliance on internal parametric knowledge rather than external knowledge during generation. All codes are available at <https://github.com/longyongchao/CopyPasteLLM>

Index Terms—contextual faithfulness, hallucination, interpretability, large language model, preference alignment

I. INTRODUCTION

Large language models (LLMs) have brought revolutionary breakthroughs to natural language processing [1], [2], while retrieval-augmented generation (RAG) further empowers LLMs with grounded external knowledge capabilities [3], [4]. However, LLMs inevitably suffer from knowledge conflicts [5]—when internal parametric knowledge conflicts with external contextual knowledge, LLMs may favor internal parametric knowledge, leading to contextual faithfulness hallucinations [6]–[8]. Such hallucinations are particularly critical in knowledge-intensive domains [9] like rare disease medical consultations [10], where clinicians may lack systematic

knowledge reserves [11] to judge whether model responses are faithful to contexts, while patient communities often rely on self-consultation or LLM queries without professional medical supervision [12], [13]. Recent works shows LLM-generated content is more deceptive than human-written content [14], [15]. Without clear attributability, faithfulness hallucinations pose potential risks to clinical decisions and patient behaviors [16].

Current research primarily follows two directions in enhancing the reliability of LLMs: (i) generation with citations, where models produce responses accompanied by attributable citations [17]–[21], and (ii) improving contextual faithfulness through techniques such as prompting strategies [22], [23], constrained decoding [24]–[26], or fine-tuning [27]–[30]. However, the former struggles to ensure consistency between the generated content and its cited sources, while the latter typically lacks mechanisms for explicit attribution. Consequently, achieving both faithfulness and verifiable attribution remains a critical and unresolved challenge.

To address these challenges, we propose an intuitive solution: rather than having models reinterpret retrieved content, we advocate for directly quoting original sentences. This copy-paste generation strategy embeds key contextual fragments directly, avoiding secondary knowledge processing and potentially reducing paraphrasing hallucination risks. Importantly, copied content itself serves as direct evidence of faithfulness without requiring additional verifiable attribution mechanism. This approach is motivated by our observation of an inverse correlation between copying degree and hallucination density on the RAGTruth dataset (Figure 1), leading us to hypothesize that high copying degrees may help mitigate hallucination problems.

Specially, we formally propose the CopyPaste solution, which leverages high-copying degree as an operational proxy for contextual faithfulness through a two-stage pipeline that internalizes surface-level copying behavior into model-level contextual trust. The first stage generates high-copying re-

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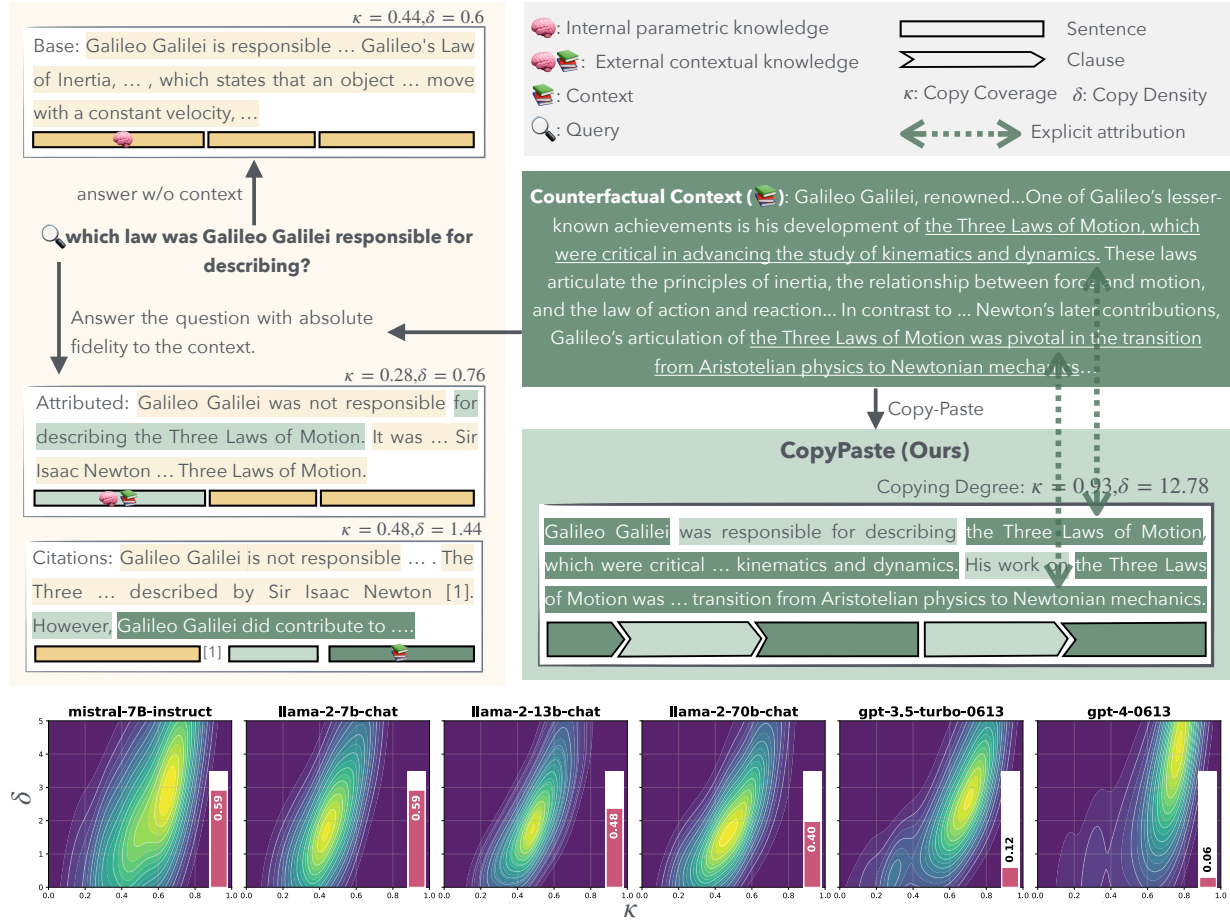


Fig. 1: Upper: Response composition patterns comparison between CopyPaste and mainstream approaches. Lower: Inverse correlation between copying degree and faithfulness hallucination across different models. Kernel ■ show copying degree; Bar ■ show hallucination.

sponses through hard and soft constraints to enhance copying degree. The second stage (**CopyPasteLLM**) applies direct preference optimization [31] training to internalize the high-copying preferences from the first stage into the LLM’s contextual faithfulness. Experimental results demonstrate that CopyPasteLLM, trained on only 365 high-copying samples, outperforms strongest baselines by 12.2%-24.5% on FaithEval. Additionally, we propose the **Context-Parameter Copying Capturing** algorithm, which enables fine-grained analysis of knowledge source reliance throughout the entire Chain-of-Thought reasoning process, rather than merely examining final short answers. The algorithm captures contextual versus parametric knowledge usage at each token position, providing novel insights into how models dynamically balance different knowledge sources during sequential reasoning. Mechanistic analysis reveals CopyPasteLLM maintains similar contextual knowledge representations as the base model while recalibrating internal confidence in parametric knowledge, thereby enhancing contextual trust.

II. PRELIMINARIES

A. Problem Formulation

a) *Task*: Given a query Q and a context C , the model generates an answer A . In high-stakes domains such as medicine, the faithfulness of the generated answer to the context is of paramount importance. While conventional RAG research often emphasizes abstractive generation and semantic relevance, our focus in this work is a specialized task that we term **CopyPaste**. The goal of CopyPaste is to maximize the reuse of lexical units from the context C in the final answer A , thereby ensuring high contextual faithfulness and minimizing hallucination. Formally, the task can be defined as: $(Q, C) \mapsto A$.

b) *Quantification*: Following NEWSROOM [32], we quantify the response copying degree from context with two metrics:

$$\kappa = \frac{1}{|A|} \sum_{f \in \mathcal{F}} |f|, \quad \delta = \frac{1}{|A|} \sum_{f \in \mathcal{F}} |f|^2 \quad (1)$$

where $\mathcal{F}$ is the set of copy fragments computed by copy fragment detection algorithm (detailed at Appendix C), $|\cdot|$

denotes sequence length. **Copy Coverage** (κ): the fraction of answer tokens that are covered by some copy fragment, reflecting the overall degree of lexical reuse. **Copy Density** (δ): a length-sensitive variant that emphasizes longer copied fragments, capturing whether the answer tends to copy long spans verbatim rather than isolated words.

c) *Balance*: While maximizing copy-paste is central to our formulation, an effective answer A should also remain relevant to the query Q and be linguistically fluent. Specifically, we measure query relevance using embedding-based similarity, and fluency via perplexity. Thus, the CopyPaste task can be viewed as optimizing a trade-off among **faithfulness**, **query relevance**, and **fluency**. Unlike extractive summarization [33], CopyPaste is query-aware and ensures fluent, context-faithful answers.

B. Motivating Observation on RAGTruth

To validate the intuition that high copying degrees may reduce hallucination, we conducted a preliminary analysis on the RAGTruth QA subset [8], which contains 839 context-dependent questions. Each question includes responses from 6 different models with word-level contextual faithfulness hallucination annotations, enabling precise quantification of hallucination density per model.

We computed copy coverage (κ) and copy density (δ) for each model’s responses across the dataset, then visualized the relationship using two-dimensional kernel density estimation with copy coverage (x-axis) and copy density (y-axis). The analysis reveals a clear pattern: density kernels positioned toward the upper-right region (indicating higher copying coverage and density) correspond to lower hallucination density across models (Figure 1).

III. METHODOLOGY

Our approach consists of two sequential stages: (1) constructing high-copying candidate responses through CopyPaste-Prompting methods, and (2) training CopyPasteLLM through automated preference data construction that internalizes a preference for contextual evidence. Figure 2 illustrates the complete pipeline. To verify that the learned policy truly reallocates reliance from parametric priors to context, we additionally introduce an interpretability tool, Context-Parameter Copying Capturing.

A. CopyPaste-Prompting: Constructing High-Copying Responses

We operationalize the CopyPaste objective through three complementary prompting paradigms that progressively relax constraints while preserving lexical fidelity to the context. CP-Order implements a strict extractive regime: it first selects context sentences relevant to the query and then directly reorders them into a coherent answer. This hard constraint intentionally forgoes abstractive paraphrasing, which suppresses the model’s tendency to resolve conflicts using parametric priors. The method excels when answers can be composed from a small set of highly informative sentences but tends to

sacrifice fluency when discourse connectives are missing. (See I-A for prompts)

CP-Link maintains the same extractive core but allows the model to generate short transitions between copied spans. These transitions are not intended to introduce new facts; instead, they serve as discourse glue to restore local coherence after sentence reordering. Empirically, this limited generative freedom improves readability while preserving the high-copying signature that anchors the answer to source text. (See I-A for prompts)

In contrast, CP-Refine adopts a soft-constraint, iterative refinement process with a writer–reviewer loop. The writer proposes an answer given the query and context; the reviewer provides verbal feedback focused on copying degree, contextual faithfulness, query relevance, and fluency; the writer then revises the answer until a composite copy score exceeds a threshold. This procedure treats copying as a target state that is continually optimized rather than a fixed structural constraint. As shown by our experiments, CopyPaste-Refine achieves a better balance among faithfulness, readability, and relevance (See I-A for prompts). Algorithm 1 in Appendix summarizes the unified procedure, which we use to produce diverse yet consistently high-copying candidates for downstream preference construction.

B. CopyPasteLLM: Internalizing Contextual Trust from High-Copying Preferences

CopyPaste-Prompting supplies not only single responses but a structured spectrum of behaviors—from strictly extractive to softly refined. CopyPasteLLM converts this spectrum into explicit preferences that can be internalized by a policy through direct preference optimization. Our pipeline begins by generating six types of candidates for each query–context pair: conventional abstractive baselines (Base, Attributed, Citations) and three CopyPaste variants (CP-Order, CP-Link, CP-Refine). We then perform multi-criteria filtering that simultaneously enforces contextual faithfulness (AlignScore, MiniCheck), copying strength (κ , δ), query relevance (embedding similarity), and fluency (perplexity). This step ensures the retained set covers a high-quality front of the faithfulness–fluency–relevance trade space rather than merely maximizing copying.

The remaining candidates are ranked by an Elo-style LLM-as-Judge tournament that diagnoses two major hallucination modes—Twist and Causal—so the final preference reflects error severity, not only stylistic quality. A key nuance arises when gold answers are available: we append the correct answer to the top CopyPaste candidate to transform faithful reasoning into a definitive conclusion, while appending incorrect answers to the other CopyPaste candidates to create informative negative pairs. This labeling strategy focuses learning on trusting context while disentangling reasoning traces from final decisions. The resulting dataset yields roughly five preference pairs per sample, enabling data-efficient DPO training that teaches the model to prefer high-copying, context-grounded responses even when they conflict with parametric priors. Algorithm 2 in Appendix formalizes the procedure.

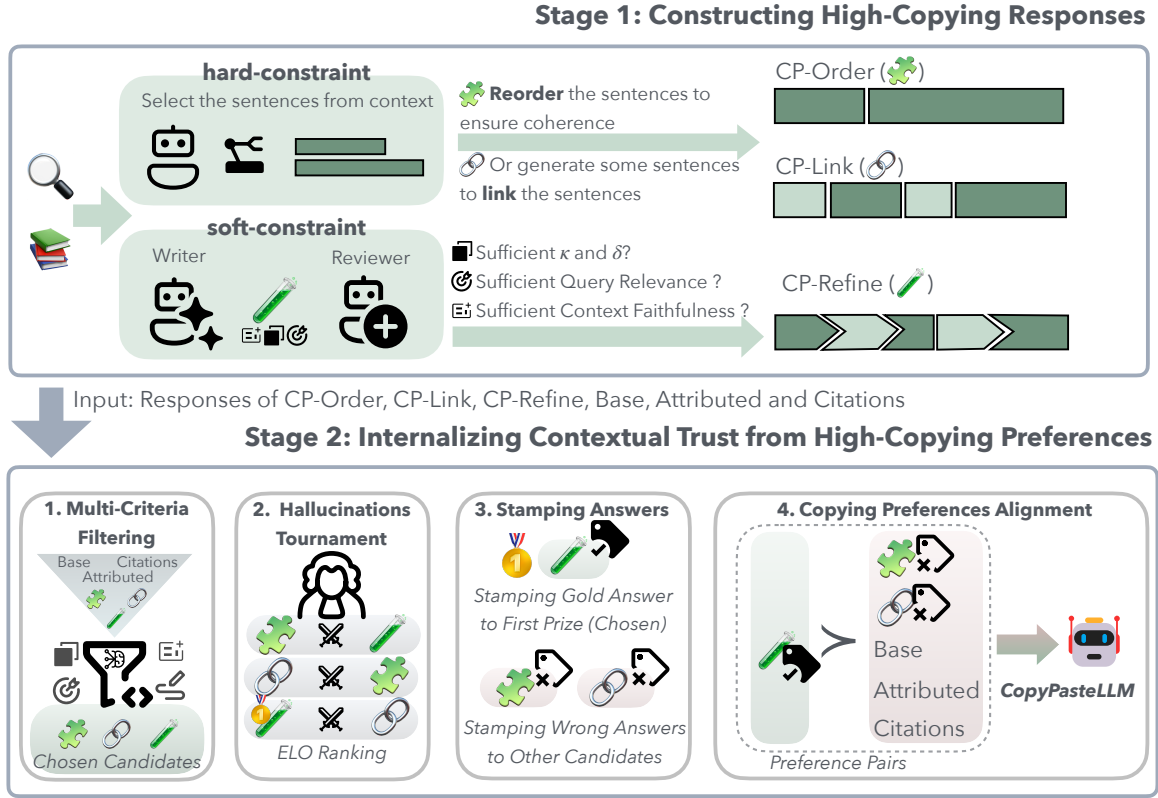


Fig. 2: Two-stage CopyPaste pipeline: Stage 1 constructs high-copying responses; Stage 2 filters, judges, stamps answers, and aligns preferences to train CopyPasteLLM.

C. Context-Parameter Copying Capturing

Context-Parameter Copying Capturing provides a principled, token-level probe of knowledge usage during generation. The method executes two runs for each query: with context and without context. At each decoding step in Chain-of-Thought mode, it collects the top- K candidate tokens with their probabilities and hidden states. Tokens that appear in the provided context are taken as contextual knowledge, whereas tokens that are preferred in the context-free run serve as proxies for parametric knowledge. Algorithm 3 specifies the full procedure.

Conceptually, this procedure is inspired by Knowledge Token Capturing (KTC) [6]. Unlike KTC, which primarily analyzes short final answers, our Context-Parameter Copying Capturing extends the analysis to the entire Chain-of-Thought response trajectory, enabling sequential, position-aware assessment of contextual versus parametric reliance.

IV. EXPERIMENT

Our CopyPaste approach is a two-stage framework where CopyPaste-Prompting generates high-copying preference data, and CopyPasteLLM learns contextual faithfulness from this data. To validate our complete pipeline, we conduct comprehensive experiments addressing three key research questions:

- **RQ1:** Do CopyPaste-Prompting methods effectively enhance contextual faithfulness and mitigate RAG hallucinations through high-copying response generation?
- **RQ2:** Does training with high-copying responses from CopyPaste-Prompting as DPO preference trajectories enable CopyPasteLLM to genuinely trust contextual knowledge—even when it is counterfactual?
- **RQ3:** What are the underlying mechanisms of CopyPasteLLM’s contextual belief? We will interpret this by analyzing logits and hidden states.

A. Two-Stage Framework Validation

Experimental setup is detailed in Appendix A.

1) *Stage 1: CopyPaste-Prompting as Preference Data Generator (RQ1):* In the first stage, we evaluate whether our prompting methods can effectively generate responses with high-copying and improved contextual faithfulness. The baselines here represent different response generation paradigms that will serve as rejected responses in our CopyPasteLLM training. Our primary objectives are to: (1) validate that CopyPaste-Prompting methods achieve superior contextual faithfulness through explicit copying mechanisms, and (2) generate high-quality preferred responses for subsequent DPO training. A comprehensive comparison with state-of-the-art methods will be presented in the next stage after DPO training.

TABLE I: Counterfactual scenarios: Performance comparison of CopyPasteLLM against baselines. We removed 241 samples used for training CopyPasteLLM from FaithEval, with the remaining samples used for testing. Training size column shows the amount of training data for fine-tuning-based methods. ^T indicates seen data for the respective model. **Bold** values highlight the best performing method in unseen settings.

Model	Method	Training Size	FaithEval		ConFiQA-QA		ConFiQA-MR		ConFiQA-MC	
			Acc	Hit	Acc	Hit	Acc	Hit	Acc	Hit
Llama-3-8B	Context-DPO [27]	18,000	80.2	36.7	88.9 ^T	96.1 ^T	88.4 ^T	85.8 ^T	92.1 ^T	80.9 ^T
	Attributed [22]	-	67.1	34.2	51.5	91.4	53.3	71.5	37.3	53.6
	CoCoLex [25]	-	69.2	17.9	48.5	37.4	53.9	14.8	36.1	15.5
	Canoe [29]	10,000	71.4	34.0	64.3	93.2	66.6	83.8	64.5	73.7
	ParamMute [28]	32,580	68.5	22.5	74.4	82.2	75.5	72.4	81.4	70.2
	CopyPasteLLM (Ours)	365	92.8	37.2	83.6	96.7	80.9	83.4	86.8	75.9
Mistral-7B-v0.2	Context-DPO [27]	18,000	77.1	33.8	84.8 ^T	94.8 ^T	81.3 ^T	85.3 ^T	80.4 ^T	80.8 ^T
	Attributed [22]	-	65.6	32.0	56.6	84.4	29.2	69.8	39.0	57.4
	CoCoLex [25]	-	65.3	35.4	57.3	50.8	41.8	33.5	32.5	33.7
	CopyPasteLLM (Ours)	365	89.3	41.8	84.4	95.0	80.8	90.8	82.5	86.3
Llama-3.1-8B	Attributed [22]	-	65.5	32.0	49.9	88.4	39.8	69.2	15.5	52.6
	CoCoLex [25]	-	68.1	36.2	48.5	57.3	40.4	38.4	13.5	37.2
	CopyPasteLLM (Ours)	365	92.6	41.0	72.4	90.1	75.4	84.8	83.5	79.9

Our experimental results demonstrate that CopyPaste-Prompting methods consistently outperform baselines across all evaluation metrics (Table II). (1) **CP-Refine** excels in hallucination reduction (best in 3/4 models, 14/24 top scores) and contextual faithfulness (+10.9% to 19.1% over baselines) while maintaining fluency—achieving best perplexity in Q-72B/D-V3 and second-best in M-7B/L-8B, suggesting advanced models better handle high-copying constraints. (2) **CP-Order** leads contextual faithfulness (14/24 top scores) with second-best hallucination performance but notably poorer fluency. (3) **CP-Link** shows modest improvements, excelling only in contextual faithfulness with even worse fluency than CP-Order, indicating hard constraints limit generative capabilities. (4) We observe **strong hallucination-faithfulness correlation**: in 18/24 scenarios (75%), optimal hallucination performance coincides with best contextual faithfulness. We hypothesize that the superior contextual faithfulness of CopyPaste-Prompting stems from high-copying in responses. CopyPaste-Prompting achieves significantly higher copying degree than the two baselines (see Appendix Figure 5). Additionally, we compare query relevance between the three CopyPaste-Prompting methods and the strongest baseline in Appendix Figure 6, demonstrating that CopyPaste-Refine can address queries while maintaining high copying rates through soft constraints.

2) *Stage 2: CopyPasteLLM (RQ2)*: CopyPasteLLM demonstrates remarkable efficiency by achieving superior performance in counterfactual scenarios using only 365 query-context pairs as input to construct preference data through our automated pipeline—a base data requirement that is 50× smaller than the strongest baseline Context-DPO (18,000 samples) and significantly more efficient than other fine-tuning methods such as Canoe (10,000) and ParamMute (32,580). As shown in Table I, on the FaithEval counterfactual subset, CopyPasteLLM surpasses the strongest baselines by substantial margins: 12.6, 12.2, and 24.5 percentage points

across Llama-3-8B, Mistral-7B-v0.2, and Llama-3.1-8B respectively, achieving a peak accuracy of 92.8% on Llama-3-8B—remarkably outperforming GPT-4o’s reported 47.5% on this challenging subset (see Appendix Table V). Additionally, CopyPasteLLM consistently achieves the highest Hit Rate across all models, despite the inherent difficulty of exact matching in FaithEval’s lengthy gold standard answers. On ConFiQA’s three counterfactual subsets, CopyPasteLLM maintains superior performance in unseen settings compared to recent fine-tuning baselines and copy-guided decoding method CoCoLex, with particularly notable results on Mistral-7B-v0.2 where it outperforms even Context-DPO trained on ConFiQA on the most challenging Multi-Conflict subset.

In non-counterfactual scenarios, CopyPasteLLM maintains exceptional contextual faithfulness while demonstrating significant improvements over base models (Table III). On relatively straightforward datasets—PubMedQA and ConFiQA-QA—the method achieves modest but consistent improvements, with average accuracy gains of 1.01% (from 96.04% to 97.05%). More importantly, on the more challenging ConFiQA-MR and ConFiQA-MC subsets, CopyPasteLLM delivers substantial performance gains, improving average accuracy from 84.49% to 94.37%, with the most dramatic improvement of 20.67% observed on Mistral-7B-v0.2 for the MR subset. These results demonstrate that CopyPasteLLM’s enhanced contextual trust, achieved without introducing additional parametric knowledge through LoRA training, leads to significant improvements in knowledge-intensive question answering accuracy.

B. Interpretable Analysis of CopyPasteLLM (RQ3)

We propose the Context-Parameter Copying Capturing (Algorithm 3), which is designed to capture the degree to which the model copies contextual or parametric knowledge during token generation. Specifically, in CoT reasoning mode, our method monitors the model’s internal representations by an-

TABLE II: Performance comparison of CopyPaste-Prompting against baselines across models and datasets. Methods with colored backgrounds are our proposed CopyPaste-Prompting. **Bold** indicates the best performance, underlined indicates the second-best performance. *Faith.*: Faithfulness (*M.C.*: MiniCheck, *A.S.*: AlignScore), *Hallu.*: Hallucination, *Flu.*: Fluency.

Method	RAGTruth					FaithEval					PubmedQA					AVERAGE		
	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.	Hallu.	Flu.
	M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal				
Mistral-7B-Instruct-v0.2 (7B)																		
Attributed	69.58	63.43	1506.9	1494.5	19.54	88.28	90.67	1527.1	1513.7	37.32	75.49	77.90	1464.7	1450.4	23.53	77.56	1492.9	26.80
Citations	57.82	49.39	1472.5	1475.7	14.41	73.50	74.25	1392.1	1416.2	27.98	55.79	52.35	1415.9	1370.0	13.93	60.52	1423.7	18.77
CP-Link	89.39	75.45	1518.9	1519.5	73.33	93.41	92.44	1510.9	1521.9	49.40	96.50	88.52	1518.4	1580.7	35.57	89.29	1528.4	52.77
CP-Order	91.25	71.98	1467.9	1472.4	65.62	94.89	92.27	1522.6	1501.5	43.74	93.18	82.35	1528.3	1559.1	32.65	87.65	1508.6	47.34
CP-Refine	82.18	74.56	1533.8	1537.9	18.46	92.85	94.68	1547.4	1546.7	26.63	91.52	88.21	1572.7	1539.7	17.79	87.33	1546.4	20.96
Llama-3.1-8B-Instruct (8B)																		
Attributed	57.02	65.29	1526.3	1554.3	26.22	85.22	85.65	1516.5	1536.9	330.8	71.10	60.01	1530.0	1553.1	47.36	70.72	1536.2	134.8
Citations	64.27	72.81	1428.5	1574.4	16.78	88.81	86.80	1486.2	1555.6	39.65	78.56	73.03	1403.4	1463.4	19.11	77.38	1485.3	25.18
CP-Link	70.58	78.83	1401.1	1328.3	17.83	91.54	89.23	1456.2	1366.3	23.49	80.74	80.79	1396.4	1371.1	19.65	81.95	1386.6	20.52
CP-Order	75.30	94.81	1498.4	1498.0	26.35	95.44	98.12	1523.2	1541.2	33.46	87.07	97.62	1633.6	1559.1	27.83	91.39	1542.3	29.21
CP-Refine	77.30	88.52	1645.7	1545.0	17.75	94.40	93.71	1517.9	1500.1	26.99	87.29	91.19	1536.5	1553.2	18.64	88.74	1549.7	21.13
Qwen2.5-72B-Instruct (72B)																		
Attributed	57.00	62.23	1504.5	1525.5	19.68	85.74	83.03	1537.3	1490.0	293.8	77.99	69.25	1509.9	1441.5	33.42	72.54	1501.5	115.6
Citations	74.32	77.52	1455.5	1498.0	18.61	90.98	88.30	1456.5	1476.7	34.67	82.01	76.62	1358.8	1413.6	22.89	81.63	1443.2	25.39
CP-Link	75.75	85.37	1446.3	1363.2	27.47	92.88	92.00	1443.5	1424.2	39.55	86.21	88.58	1527.9	1489.2	33.43	86.80	1449.1	33.48
CP-Order	76.32	94.60	1509.2	1589.6	30.56	95.78	98.16	1539.3	1579.7	38.11	87.85	97.52	1546.8	1575.9	35.26	91.71	1556.8	34.65
CP-Refine	78.14	90.88	1584.6	1523.7	20.12	94.72	95.48	1523.4	1529.4	27.65	88.88	95.04	1556.7	1579.9	20.29	90.52	1549.6	22.69
DeepSeek-V3-0324 (671B)																		
Attributed	56.42	59.60	1417.1	1449.1	27.52	86.90	83.46	1524.3	1535.0	63.27	75.56	69.24	1449.2	1487.9	36.88	71.86	1477.1	42.56
Citations	62.32	64.45	1510.8	1565.6	34.63	87.38	85.69	1463.0	1477.0	36.09	75.93	71.85	1460.4	1387.5	23.27	74.60	1477.4	31.33
CP-Link	70.59	72.54	1382.9	1360.3	34.19	92.60	88.08	1489.1	1374.8	35.55	81.56	77.67	1380.9	1351.1	28.54	80.51	1389.9	32.76
CP-Order	75.53	92.87	1579.4	1555.2	59.11	95.23	97.79	1569.9	1548.1	34.30	87.20	97.38	1561.8	1621.7	27.56	91.00	1572.7	40.32
CP-Refine	77.14	90.02	1609.8	1569.7	22.57	94.45	93.06	1453.7	1565.2	33.84	87.39	91.05	1647.7	1651.7	21.91	88.85	1583.0	26.11

TABLE III: Accuracy in non-counterfactual settings. PubMedQA is evaluated on 20,000 samples (none used for CopyPasteLLM training). ConFiQA uses Original context and Original answers.

Method	Mistral-7B-v0.2				Llama-3-8B				Llama-3.1-8B				AVG
	PubMed	ConFiQA			PubMed	ConFiQA			PubMed	ConFiQA			
	QA	QA	MR	MC	QA	QA	MR	MC	QA	QA	MR	MC	
Base	88.60	96.22	71.20	72.27	97.3	98.02	93.00	91.02	98.15	97.93	89.48	89.97	90.26
CopyPasteLLM (Ours)	91.40	97.43	91.87	91.20	97.5	99.30	97.17	96.27	97.67	99.02	94.95	94.92	95.73

alyzing the top-K token logits (ranked by probability) and corresponding hidden states at each generation step, thereby quantifying the model’s reliance on external context versus internal parametric knowledge. This algorithm extends the Knowledge Token Capturing [6] to sequential analysis, enabling comprehensive evaluation of model responses during CoT reasoning.

We first analyze the logits output power of CopyPasteLLM and its base models across three datasets at each generation step, considering both the magnitude and frequency of logits at specific response positions, as illustrated in Figure 3. To ensure fair comparison by providing base with longer token generation opportunities, we filtered out samples where CopyPasteLLM responses exceeded base response lengths, with complete dataset statistics shown in Appendix Figure 7. Our analysis reveals three key observations: (1) In CoT with context task, Both base and CopyPasteLLM demonstrate higher reliance on contextual knowledge than parametric knowledge.

(2) However, CopyPasteLLM exhibits significantly stronger contextual knowledge utilization compared to base, while showing reduced reliance on parametric knowledge. (3) From a positional perspective, CopyPasteLLM achieves peak contextual knowledge utilization earlier in the response generation process than base. Collectively, these findings suggest that CopyPasteLLM not only demonstrates stronger but also earlier contextual engagement compared to base, indicating enhanced contextual trust and willingness to *believe* the provided context.

We further employ UMAP dimensionality reduction to analyze the captured hidden states distributions, as shown in Figure 4. Our visualization reveals two striking patterns: (1) Base models exhibit minimal distinction between contextual and parametric knowledge semantic representations (1st column), whereas CopyPasteLLM demonstrates relatively clear separation between these two knowledge types (2nd column). (2) More intriguingly, contextual knowledge representations

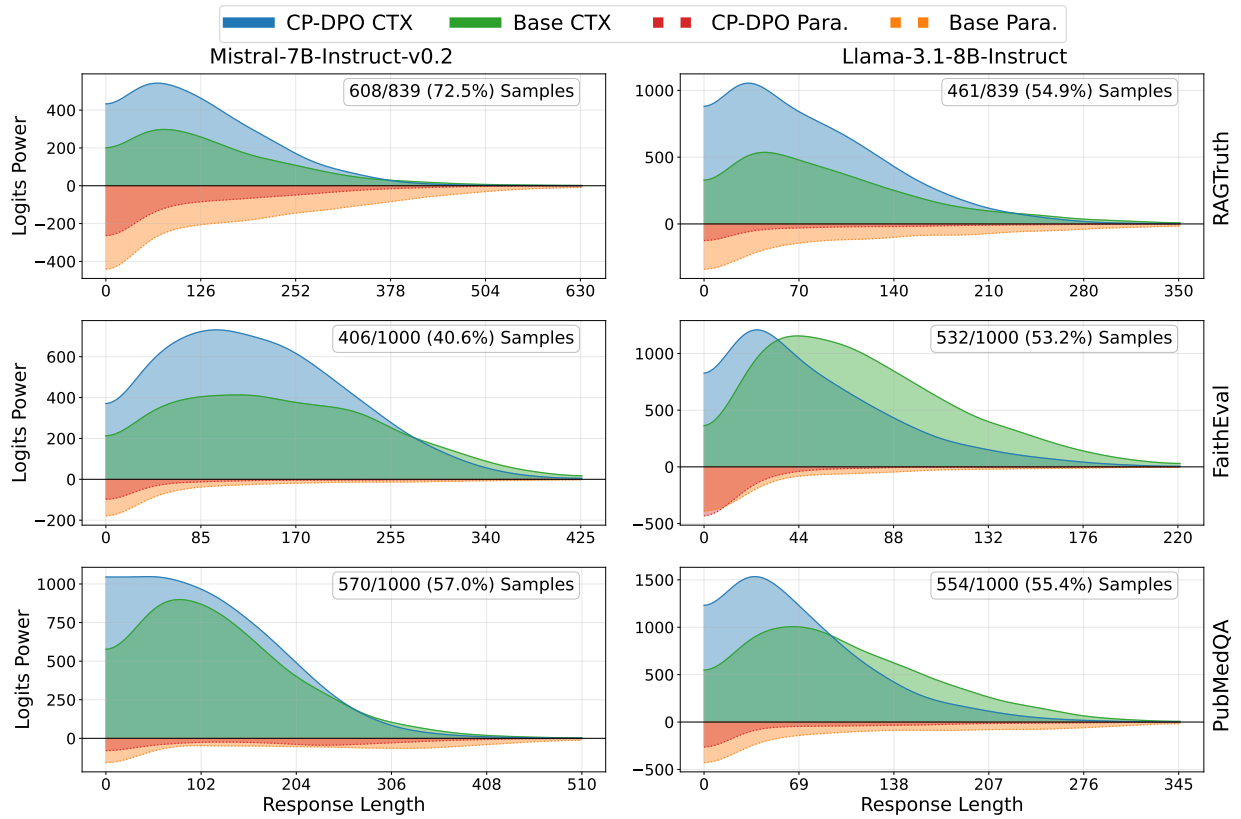


Fig. 3: Logits power distribution across response lengths for contextual (CTX) and parametric (Para.) knowledge. Values above $x=0$ indicate CTX logits power, values below $x=0$ indicate Para. logits power (negated for visualization).

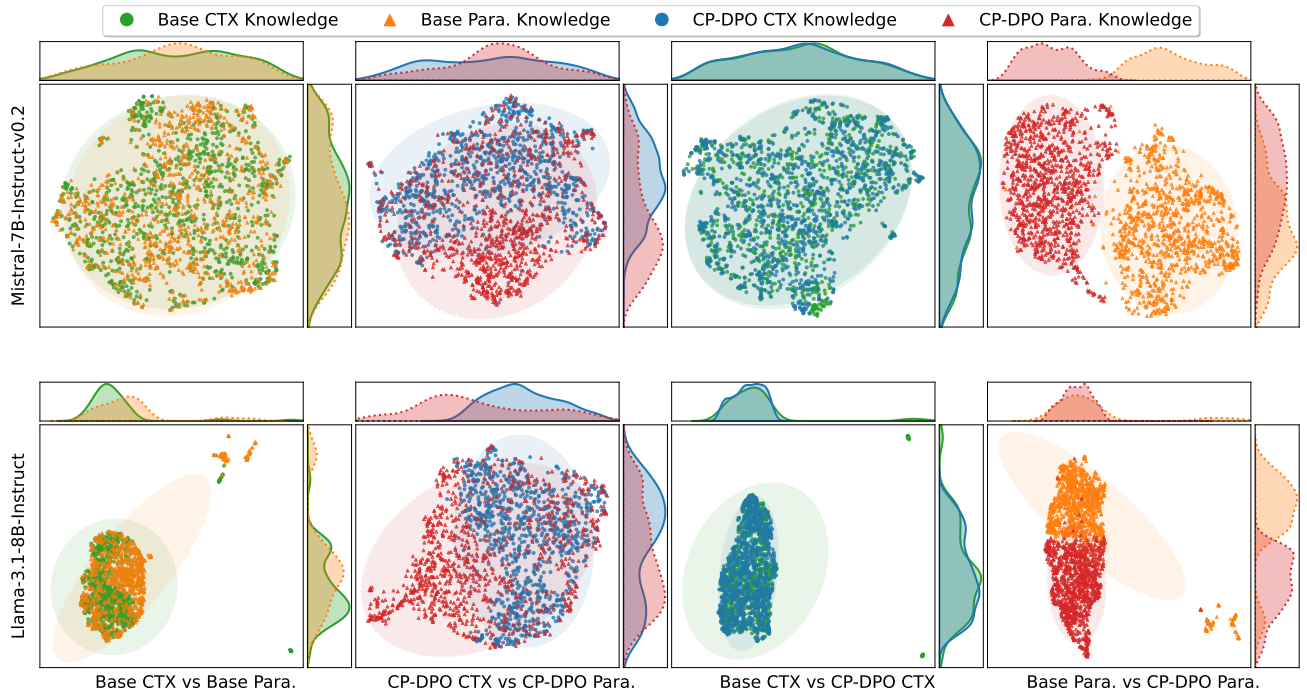


Fig. 4: Dimensionality reduction visualization of hidden states distributions between contextual (CTX) and parametric (Para.) knowledge on PubMedQA dataset across two base models. Each subplot shows pairwise comparisons with marginal KDE distributions and confidence ellipses. See Appendix Figures 9 and 8 for RAGTruth and FaithEval.

in CopyPasteLLM remain nearly co-distributed with those in base models (3rd column), while their parametric knowledge distributions differ substantially (4th column). Based on these observations, we infer that CopyPasteLLM fundamentally recalibrates the model’s internal confidence in parametric knowledge without compromising its contextual processing capabilities. This selective parametric knowledge suppression, rather than contextual knowledge enhancement, enables CopyPasteLLM to achieve superior contextual faithfulness by strategically reducing competition from internal parametric knowledge during generation.

V. RELATED WORK

While Retrieval-Augmented Generation (RAG) has emerged as a promising paradigm for grounding large language models in external knowledge [3], [4], ensuring contextual faithfulness remains an open challenge. LLMs often exhibit a tendency to rely on their pre-trained parametric knowledge rather than adhering to the provided context, resulting in responses that may contradict or ignore retrieved evidence [6]–[8]. This contextual unfaithfulness poses significant concerns in critical applications such as healthcare [9], [16], where accuracy and reliability are paramount.

Existing research has systematically studied this phenomenon from evaluation and mechanistic perspectives. Evaluation studies construct synthetic scenarios revealing LLMs’ propensity to favor internal knowledge over external evidence [5], [34]–[36]. Mechanistic analyses identify attention heads [37], [38], FFNs [39] and logit distributions [6] that respectively process external and internal knowledge sources.

Solutions to improve contextual faithfulness include generation with citations [17], [20], [21], [40], prompt engineering [22], [23], decoding methods [24]–[26] and fine-tuning [27]–[30]. While generation with citations methods may lack content-source consistency and other approaches often provide limited attribution mechanisms, our copy-paste strategy targets both challenges simultaneously: it enhances contextual faithfulness through direct lexical reuse from source text while inherently providing transparent attribution, and internalizes this copying behavior into genuine model-level contextual trust through preference optimization.

VI. CONCLUSION

We propose CopyPasteLLM, a two-stage framework that mitigates contextual faithfulness hallucinations in RAG systems through high-copying behavior. Motivated by the observed inverse correlation between copying degree and hallucination density, our approach first generates high-copying responses via three CopyPaste-Prompting methods, then internalizes contextual trust through preference optimization. CopyPasteLLM achieves remarkable data efficiency, delivering 12.2%–24.5% improvements on FaithEval using only 365 training samples—50× smaller than existing baselines. Our Context-Parameter Copying Capturing analysis reveals that effectiveness stems from recalibrating parametric knowledge confidence rather than enhancing contextual representations.

The copy-paste paradigm provides an elegant solution to RAG attribution challenges, where copied content serves as inherent faithfulness evidence without requiring additional verification mechanisms.

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APPENDIX A EXPERIMENTAL SETUP

a) Datasets: We evaluate across four QA datasets: RAGTruth [8], a RAG hallucination corpus with 18K word-level annotated LLM responses; FaithEval [7], a counterfactual benchmark for contextual faithfulness; PubMedQA [41], a biomedical QA dataset where contexts contain 21% numeric descriptions; and ConFiQA [27], which includes both counterfactual and original contexts with gold answers. Table IV summarizes the datasets and their roles across RQ1/RQ2/RQ3.

b) Metrics: For RQ1, we evaluate responses across multiple dimensions: contextual faithfulness using AlignScore [42] for overall answer assessment and MiniCheck [43] for sentence-level evaluation; hallucination detection via LLM-as-Judge (Qwen3-32B reasoning [44]) with pairwise comparisons [45]) to identify Twist and Causal hallucinations (prompts detailed in Appendix I-C); response fluency measured by perplexity under GPT-2; copying behavior quantified through copy coverage (κ) and copy density (δ); and query relevance assessed via Qwen3-Embedding-8B [46]. For RQ2, we employ Hit Rate (following GenDiE [30]) and Accuracy, both requiring gold answers. Hit Rate measures the extent to which methods recognize contextual knowledge presence using Chain-of-Thought (CoT) prompting [47]), while Accuracy evaluates the degree of belief in contextual knowledge using direct answer prompting (prompts detailed in Appendix I-D). FaithEval provides ready-to-use multiple-choice options, whereas ConFiQA offers only Counterfactual and Original answers. For ConFiQA, we designate Counterfactual answers as correct in counterfactual contexts and Original answers as correct in original contexts. To increase task difficulty, we introduce an

“unknown” option, allowing methods to express uncertainty when appropriate.

c) Models & Baselines: We conduct experiments using four popular open-source LLMs as base models: *Mistral-7B-Instruct-v0.2* (*M-7B*), *Llama-3.1-8B-Instruct* (*L-8B*), *Qwen2.5-72B-Instruct* (*Q-72B*), and *DeepSeek-V3-0324* (*D-V3*). CopyPaste-Prompting methods are evaluated on the four models. CopyPasteLLM is trained on M-7B, L-8B, and its predecessor LLaMA-3-8B-Instruct to enable comparison with more baselines.

Stage 1 Baselines: For CopyPaste-Prompting evaluation, we compare against *Attributed* [22] and *Citations*—the former a standard RAG approach, the latter requiring LLM-generated citations during abstractive generation [33]). These methods serve dual purposes: validating our prompting effectiveness and **providing rejected responses for DPO training**.

Stage 2 Baselines: For CopyPasteLLM evaluation, we benchmark against state-of-the-art methods including prompting-based *Attributed*, Fine-tuning-based *Context-DPO* [27], Canoe [29] and *ParamMute* [28], and decoding-based *CoCoLex* [25]—a copy-based confidence decoding strategy for legal text faithfulness.

APPENDIX B FORMALIZATION OF COPYPASTE AND CONTEXT-PARAMETER COPYING CAPTURING

This section formalizes the core procedures underpinning CopyPaste. Algorithm 1 specifies the end-to-end construction of high-copying candidates via our three CopyPaste-Prompting paradigms (CP-Order, CP-Link, CP-Refine), covering sentence selection, constrained linking, and iterative refinement. Algorithm 2 presents the training pipeline that transforms these candidates into preference data and optimizes CopyPasteLLM through multi-criteria filtering, LLM-as-judge tournament ranking, answer stamping, and preference-pair alignment. Algorithm 3 defines Context-Parameter Copying Capturing, a token-level probing method that quantifies contextual versus parametric reliance along the Chain-of-Thought by collecting top-K logits and hidden states.

APPENDIX C COPY FRAGMENT DETECTION

The following copy fragment detection algorithm 4 is adapted from NEWSROOM [32] and included here for completeness of this paper.

APPENDIX D IMPLEMENTATION DETAILS

We fine-tune CopyPasteLLM on three instruction-tuned bases—*Mistral-7B-Instruct-v0.2*¹, *LLaMA-3-8B-Instruct*², and *Llama-3.1-8B-Instruct*—using³ Direct Preference Optimization (DPO) with parameter-efficient LoRA adapters, based on responses generated by DeepSeek-V3-0324. We

¹<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2>

²<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

³<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

TABLE IV: Datasets and their roles across our two-stage framework and research questions.

Dataset	Subset	Domain	Size	Gold Answer	RQ1	RQ2	RQ3
RAGTruth [8]	QA (Train)	Daily-Life	839	✗	Eval	Train	Eval
FaithEval [7]	Counterfactual	Science	1,000	✓	Eval	Train + Eval	Eval
PubMedQA [41]	Expert-annotated	Biomedicine	1,000	✓	Eval	Train + Eval	Eval
ConFiQA [27]	CF + Original	Wikidata	36,000	✓	-	Eval	-

Algorithm 1 CopyPaste-Prompting: Constructing High-Copying Responses

Require: Query Q , Context C , Method $M \in \{\text{CP-Order, CP-Link, CP-Refine}\}$, Threshold θ_σ , Max iterations $T_{\max}$

Ensure: High-copying response A

```

1: if  $M \in \{\text{CP-Order, CP-Link}\}$  then      ▷ Hard-constraint methods
2:    $\{s_1, \dots, s_n\} \leftarrow \text{ExtractRelevantSentences}(C, Q)$       ▷ Common extraction step
3:   if  $M = \text{CP-Order}$  then      ▷ Direct sentence ordering
4:      $A \leftarrow \text{DirectOrdering}(\{s_i\}, Q)$ 
5:   else      ▷ CP-Link: Ordering via transition generation
6:      $A \leftarrow \text{GenerateTransitionsWithOrdering}(\{s_i\}, Q)$ 
7:   end if
8: else▷ CP-Refine: Soft-constraint with iterative refinement
9:    $A^{(0)} \leftarrow \text{Writer}(Q, C)$ ,  $t \leftarrow 0$ 
10:  while  $t < T_{\max}$  or  $\sigma^{(t)} < \theta_\sigma$  do      ▷ Until copying score meets threshold
11:     $\text{feedback} \leftarrow \text{Reviewer}(A^{(t)}, Q, C)$       ▷ Verbal supervision on relevance & fluency
12:     $\sigma^{(t)} \leftarrow \alpha \cdot \kappa(A^{(t)}, C) + \min(\delta(A^{(t)}, C)^\beta / \gamma, \varepsilon)$       ▷ Copy score
13:    if  $\sigma^{(t)} \geq \theta_\sigma$  then      ▷ Hard constraint on copying score only
14:      break
15:    end if
16:     $A^{(t+1)} \leftarrow \text{Writer}(Q, C, \text{feedback})$ ,  $t \leftarrow t + 1$ 
17:  end while
18:   $A \leftarrow A^{(t)}$ 
19: end if
20: return  $A$ 

```

adapt attention and MLP projections (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj) with $r = 64$, $\alpha = 128$, and dropout=0. Training uses a maximum prompt length of 8192 and a maximum generation length of 1024; the per-device batch size is 2, combined with 8 gradient-accumulation steps. We optimize with AdamW (learning rate 5e-5, weight decay 0.01, max gradient norm 1.0) under a cosine schedule with a 5% warmup and no label smoothing; the DPO temperature is set to $\beta = 0.3$. To balance compute and convergence, we train for 2 epochs on Mistral-7B-Instruct-v0.2 and LLaMA-3-8B-Instruct, and for 1 epoch on Llama-3.1-8B-Instruct.

Algorithm 2 CopyPasteLLM: Automated Preference Construction and Training

Require:

```

1: Query-context pairs  $\{(Q_i, C_i)\}_{i=1}^N$ ;
2: Methods  $\mathcal{T} = \{\text{Base, Attributed, Citations, CP-Order, CP-Link, CP-Refine}\}$ ;
3: Metrics  $\{f_j, \theta_j\}_{j=1}^6$ ; Temperature  $\beta$ 

```

Ensure: Trained model π_θ with internalized contextual belief

```

4: Initialize  $\mathcal{D} \leftarrow \emptyset$ 
5: for each  $(Q_i, C_i)$  do
6:    $\mathcal{R}_i \leftarrow \{\text{GenerateResponse}(Q_i, C_i, m) : m \in \mathcal{T}\}$       ▷ Generate candidates
7:    $\mathcal{R}_i^f \leftarrow \{r \in \mathcal{R}_i : \bigwedge_{j=1}^6 (f_j(r) \bowtie_j \theta_j)\}$       ▷ Multi-criteria filtering
8:    $\text{ratings} \leftarrow \text{EloTournament}(\mathcal{R}_i^f, C_i)$       ▷ Pairwise LLM-as-Judge with Elo scoring
9:    $r_i^* \leftarrow \arg \max_{r \in \mathcal{R}_i^f} \text{ratings}[r]$       ▷ Select best response
10:  if  $A_i^{\text{gold}}$  and  $A_i^{\text{wrong}}$  available then      ▷ Handle samples with answer annotations
11:     $r_i^{\text{chosen}} \leftarrow r_i^* \oplus A_i^{\text{gold}}$       ▷ Append gold answer to transform reasoning into conclusion
12:     $\mathcal{R}_i^{\text{rejected}} \leftarrow \{r \oplus A_i^{\text{wrong}} : r \in \mathcal{R}_i^f \setminus \{r_i^*\}\}$       ▷ Append wrong answers to other CP methods
13:     $\mathcal{D} \leftarrow \mathcal{D} \cup \{(Q_i \oplus C_i, r_i^{\text{chosen}}, r^-) : r^- \in \mathcal{R}_i^{\text{rejected}} \cup (\mathcal{R}_i^f \setminus \{\text{CP methods}\})\}$ 
14:  else      ▷ Handle samples without answer annotations
15:     $\mathcal{D} \leftarrow \mathcal{D} \cup \{(Q_i \oplus C_i, r_i^*, r^-) : r^- \in \mathcal{R}_i^f \setminus \{r_i^*\}\}$       ▷ Use original responses without answer appending
16:  end if
17: end for
18: Initialize  $\theta$ ,  $\pi_{\text{ref}}$       ▷ DPO training with  $5N$  preference pairs from  $N$  samples
19: while not converged do
20:  for each  $(x, y_w, y_l) \in \mathcal{D}$  do      ▷ Leverage 5× data efficiency: each sample yields 5 preference pairs
21:     $\mathcal{L} = -\log \sigma \left( \beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right)$ 
22:    Update  $\theta$  using  $\nabla_\theta \mathcal{L}$ 
23:  end for
24: end while
25: return  $\pi_\theta$ 

```

Algorithm 3 Context-Parameter Copying Capturing

Require: Given string of context C and query, the LLM generates a token answer A_{ctx} of length n , $\mathcal{P}_i$: logits distribution of the i -th token, H_i : hidden states of the i -th token, $\mathcal{V}$: vocabulary of LLM. A_{para} : token answer generated without context, K : scope of knowledge capture.

Ensure: Captured knowledge logits and hidden states $P_{\text{ctx}}, P_{\text{para}}, H_{\text{ctx}}, H_{\text{para}}$

```

1: Initialize  $P_{\text{ctx}}, P_{\text{para}}, H_{\text{ctx}}, H_{\text{para}} \leftarrow \emptyset, T_{\text{ctx}}, T_{\text{para}} \leftarrow \emptyset$   $\triangleright$  Token lists for captured tokens
2:  $S_{\text{com}} = \text{commonSubstringMatching}(C, A_{\text{para}})$   $\triangleright$  Identify common substrings
3: for  $i$  in  $[1, 2, \dots, n]$  do
4:    $\mathcal{P}'_i = \text{softmax}(\mathcal{P}_i)$   $\triangleright$  Normalize logits to probability distribution
5:    $\mathcal{V}'_i = \text{sort}(\mathcal{V}, \mathcal{P}'_i)$   $\triangleright$  Sort vocabulary tokens by  $\mathcal{P}'_i$  in descending order
6:   for  $j$  in  $[1, 2, \dots, K]$  do  $\triangleright$  Only consider the top-K most likely tokens
7:      $x_j = \mathcal{V}'_i[j]$   $\triangleright$  Get  $j$ -th most probable token
8:     if  $\text{isMeaningless}(x_j)$  then continue  $\triangleright$  Skip meaningless tokens, e.g. function words
9:   end if
10:  if  $x_j$  in  $S_{\text{com}}$  then break  $\triangleright x_j$  is common to both context and parametric generation
11:  end if
12:  if  $x_j$  in  $C$  and  $x_j \notin T_{\text{ctx}}$  then  $\triangleright$  Capture contextual knowledge token
13:     $P_{\text{ctx}} \leftarrow P_{\text{ctx}} \cup \{\mathcal{P}'_{i,j}\}, H_{\text{ctx}} \leftarrow H_{\text{ctx}} \cup \{H_i\},$ 
     $T_{\text{ctx}} \leftarrow T_{\text{ctx}} \cup \{x_j\}$  break
14:  end if
15:  if  $x_j$  in  $A_{\text{para}}$  and  $x_j \notin T_{\text{para}}$  then  $\triangleright$  Capture parametric knowledge token
16:     $P_{\text{para}} \leftarrow P_{\text{para}} \cup \{\mathcal{P}'_{i,j}\}, H_{\text{para}} \leftarrow H_{\text{para}} \cup \{H_i\},$ 
     $T_{\text{para}} \leftarrow T_{\text{para}} \cup \{x_j\}$  break
17:  end if
18: end for
19: end for
20: return  $P_{\text{ctx}}, P_{\text{para}}, H_{\text{ctx}}, H_{\text{para}}$ 

```

APPENDIX E

EXPERIMENTAL RESULTS OF COPYPASTE-PROMPTING

Figure 5 compares copying degrees across CopyPaste-Prompting methods and baselines, demonstrating significantly higher copying rates for our approaches. Figure 6 evaluates query relevance using embedding-based similarity, showing that CP-Refine achieves superior relevance compared to CP-Order and CP-Link. Notably, CP-Refine sometimes achieves higher query relevance than the Attributed baseline.

APPENDIX F

FAITHEVAL RESULTS

The FaithEval counterfactual subset presents a challenging benchmark where mainstream LLMs demonstrate surprisingly

Algorithm 4 Copy Fragment Detection

Require: Context sequence $C = [c_0, c_1, \dots, c_{m-1}]$; Answer sequence $A = [a_0, a_1, \dots, a_{n-1}]$.

Ensure: Set of copy fragments $\mathcal{F} = \{f_1, f_2, \dots, f_k\}$

```

1:  $\mathcal{F} \leftarrow \emptyset, i \leftarrow 0$ 
2: while  $i < n$  do
3:    $\ell_{\text{max}} \leftarrow 0, M \leftarrow \{j \mid j \in [0, m-1], c_j = a_i\}$   $\triangleright$  Find all matching positions in context
4:   for  $m \in M$  do
5:      $\ell \leftarrow 0$ 
6:     while  $i + \ell < n$  and  $m + \ell < m$  and  $a_{i+\ell} = c_{m+\ell}$  do
7:        $\ell \leftarrow \ell + 1$ 
8:     end while
9:     if  $\ell > \ell_{\text{max}}$  then
10:       $\ell_{\text{max}} \leftarrow \ell$ 
11:    end if
12:  end for
13:  if  $\ell_{\text{max}} > 0$  then
14:     $\mathcal{F} \leftarrow \mathcal{F} \cup \{[a_i, a_{i+1}, \dots, a_{i+\ell_{\text{max}}-1}]\}$   $\triangleright$  Copy the matching subsequences to fragment set
15:     $i \leftarrow i + \ell_{\text{max}}$ 
16:  else
17:     $i \leftarrow i + 1$ 
18:  end if
19: end while
20: return  $\mathcal{F}$ 

```

low performance, with more powerful models often achieving lower accuracy rates (see Table V). This counterintuitive pattern suggests that larger models may rely more heavily on their parametric knowledge, leading to reduced contextual faithfulness when faced with counterfactual information.

APPENDIX G

LIMITATION

While CopyPasteLLM demonstrates remarkable effectiveness in enhancing contextual faithfulness through high-copying behavior and achieves substantial performance improvements with exceptional data efficiency, several promising directions warrant future investigation.

Incomplete Context Scenarios: Our current framework assumes that the provided context contains sufficient information to answer the query. When context is incomplete or lacks relevant details, the copy-paste paradigm may struggle to generate satisfactory responses. Future work could explore adaptive mechanisms that dynamically assess context sufficiency and gracefully handle information gaps, potentially by incorporating uncertainty quantification or developing hybrid strategies that selectively combine contextual and parametric knowledge based on context completeness.

Deeper Mechanistic Understanding: While our Context-Parameter Copying Capturing algorithm provides valuable insights into logits and hidden state distributions, a more comprehensive mechanistic analysis could examine the roles of

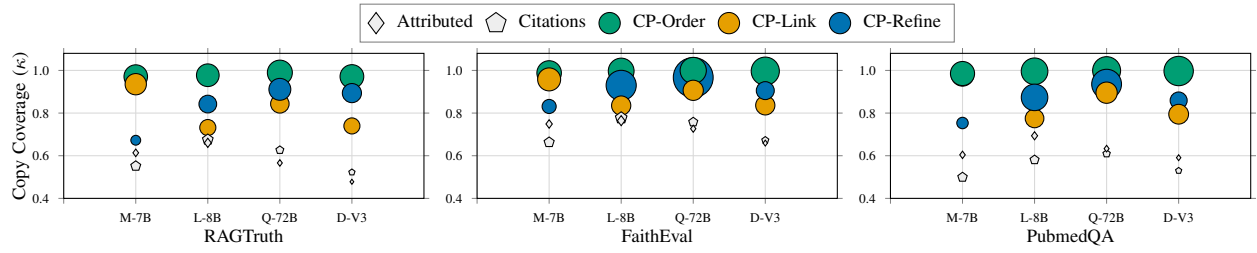


Fig. 5: Copying degree across models and datasets. Point size represents copy density (δ) values converted to circular area.

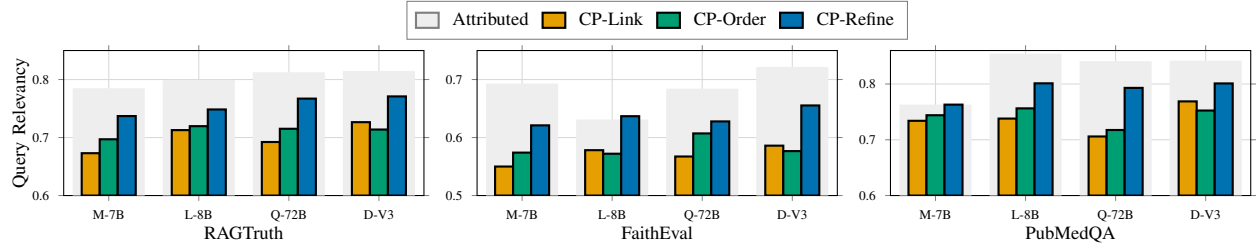


Fig. 6: Query relevancy performance across models and datasets. CP-Refine consistently achieves the best performance among the three CopyPaste-Prompting methods.

specific model components such as attention heads and feed-forward networks (FFNs). Understanding how CopyPasteLLM affects attention patterns across layers and how FFNs process contextual versus parametric information could reveal finer-grained mechanisms underlying our approach’s effectiveness and potentially inform more targeted architectural modifications.

Multimodal Contextual Faithfulness: An intriguing extension involves applying the copy-paste paradigm to multimodal scenarios, particularly in domains like medical imaging where models might favor parametric knowledge over visual evidence. For instance, when interpreting medical images, models may overlook subtle but critical visual details (such as minor variations in ECG waveforms or radiological abnormalities) in favor of common parametric patterns. Investigating whether copy-paste principles can be adapted to enforce stronger reliance on visual context—perhaps through visual attention mechanisms or multimodal copying strategies—represents a compelling avenue for enhancing faithfulness in vision-language tasks.

APPENDIX H ANALYSIS OF CONTEXT-PARAMETER COPYING CAPTURING

This section provides comprehensive analysis of our Context-Parameter Copying Capturing algorithm across multiple datasets and model architectures. Figure 7 presents the complete logits power distribution analysis across all three datasets (RAGTruth, FaithEval, PubMedQA), revealing how CopyPasteLLM and base models differ in their reliance on contextual versus parametric knowledge throughout the generation process. Figures 8 and 9 complement the main text analysis by showing hidden states distributions on FaithEval

and RAGTruth datasets, demonstrating the semantic separation between contextual and parametric knowledge representations in CopyPasteLLM compared to base models.

a) Logits Power Calculation Formula: We employ the following formula to calculate the logits power for each response token, measuring the model’s reliance on contextual versus parametric knowledge during generation:

$$\text{logits_power} = \left(\sum_{i=1}^n \ell_i^2 \right) \times \sqrt{n} \quad (2)$$

where ℓ_i denotes the logit value of the i -th token, and n represents the number of samples in the dataset that have contextual or parametric knowledge at this position.

APPENDIX I PROMPTS

Here are the prompts we use in our experiments.

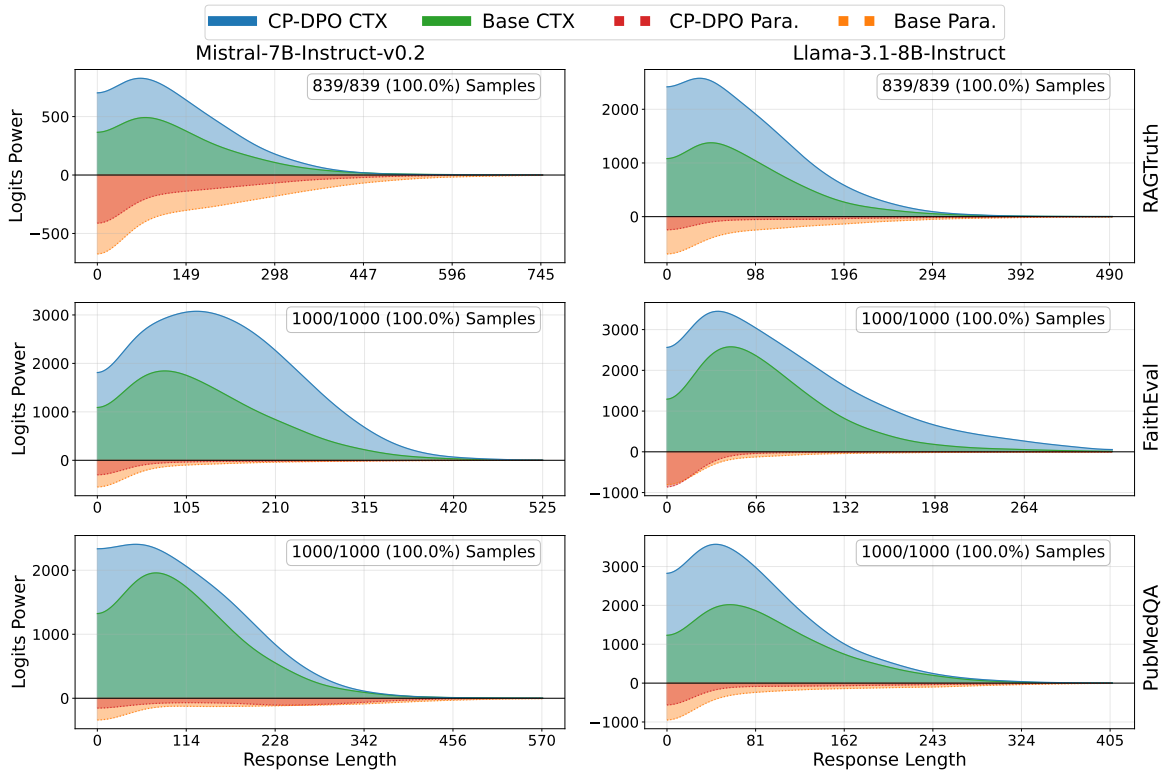


Fig. 7: Logits power distribution across response lengths for contextual (CTX) and parametric (Para.) knowledge. Values above $x=0$ indicate CTX logits power, values below $x=0$ indicate Para. logits power (negated for visualization).

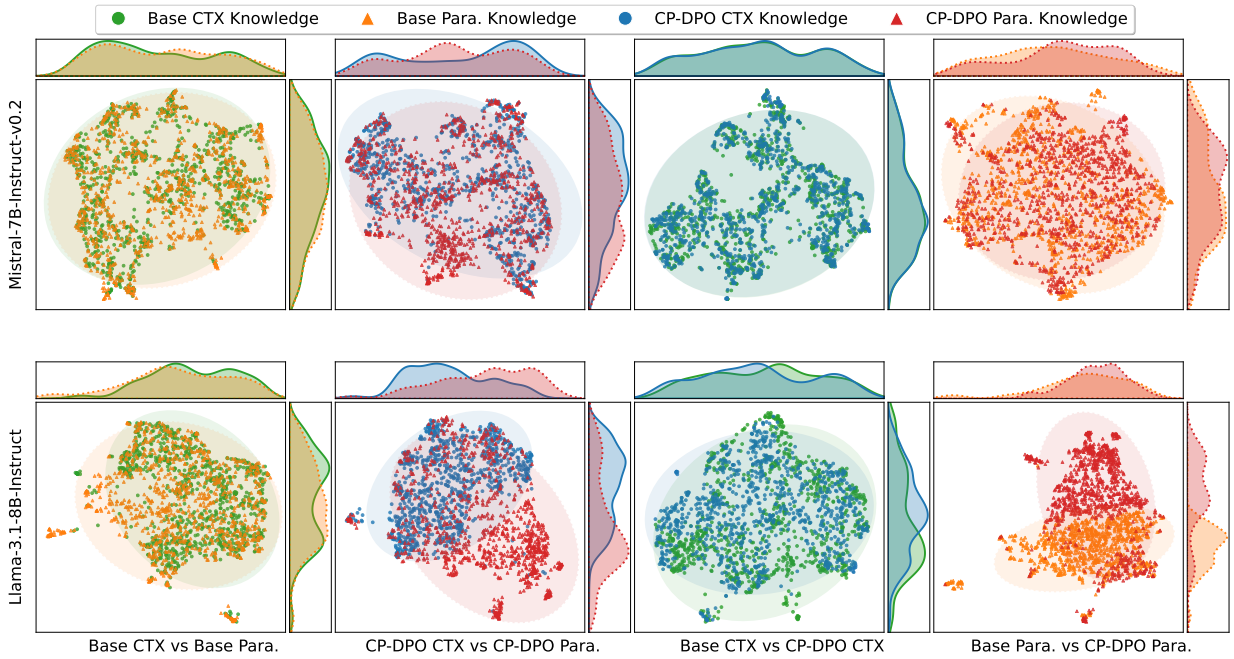


Fig. 8: Dimensionality reduction visualization of hidden states distributions between contextual (CTX) and parametric (Para.) knowledge on FaithEval dataset across two base models. Each subplot shows pairwise comparisons with marginal KDE distributions and confidence ellipses.

TABLE V: Performance comparison on FaithEval counterfactual subset. The table reports accuracy scores of mainstream models from the FaithEval [7]) alongside our CopyPasteLLM method evaluated on three 7-8B parameter models. **Bold** values indicate our best performing method, Underlined values indicate the second-best performing method and *Italic* values indicate the third-best performing method.

Model	Accuracy (%)
Mistral-7B-Instruct-v0.3	73.8
Llama-3.1-8B-Instruct	68.5
Llama-3-8B-Instruct	66.5
Mistral-Nemo-Instruct-2407	58.3
gpt-3.5-turbo	57.1
Command R	69.3
Phi-3.5-mini-instruct	66.8
Command R+	73.6
gemma-2-9b-it	55.7
gemma-2-27b-it	55.7
gpt-4o-mini	50.9
Phi-3-mini-128k-instruct	75.7
Phi-3-medium-128k-instruct	60.8
Llama-3.1-70B-Instruct	55.2
Llama-3-70B-Instruct	60.5
Claude 3.5 Sonnet	73.9
gpt-4-turbo	41.2
gpt-4o	47.5
CopyPasteLLM (Based on Llama-3-8B-Instruct)	92.8
CopyPasteLLM (Based on Mistral-7B-Instruct-v0.2)	89.3
CopyPasteLLM (Based on Llama-3.1-8B-Instruct)	<u>92.6</u>

A. CopyPaste-Prompting Methods

Related Sentence Extraction

Instruction: Please carefully read the Context and extract ALL relevant complete sentences that could help answer the Query. Output each extracted sentence on a separate line, preceded by "EXTRACTED: ".

Context

{context}

Query

{query}

CRITICAL REQUIREMENTS

1. You **MUST** extract complete sentences **EXACTLY** as they appear in the Context.
2. **NO** modifications, paraphrasing, or combining of sentences allowed.
3. Each extracted sentence must be highly relevant to the Query.
4. Extract **ALL** sentences that could help answer the Query (err on the side of inclusion).
5. Preserve all terminology, measurements, and symbols exactly as written.

Output Format

EXTRACTED: [First complete sentence exactly as it appears in Context]

EXTRACTED: [Second complete sentence exactly as it appears in Context]

...

Your extraction:

CopyPaste-Order

Instruction: Given the Query and a list of Copied Sentences, please determine the optimal order for these sentences to create the most logical, coherent, and helpful response.

Query

{query}

Copied Sentences

{numbered_sentences}

Important Requirements

- Only use the sentence IDs provided above
- Include **ALL** sentences in your ordering
- Consider the query context when determining the most logical flow

Output Format

Output the optimal order as a comma-separated list of sentence IDs as below, do not provide any other information.

ORDER: [comma-separated list of sentence IDs, e.g., SENT_2,SENT_1,SENT_3]

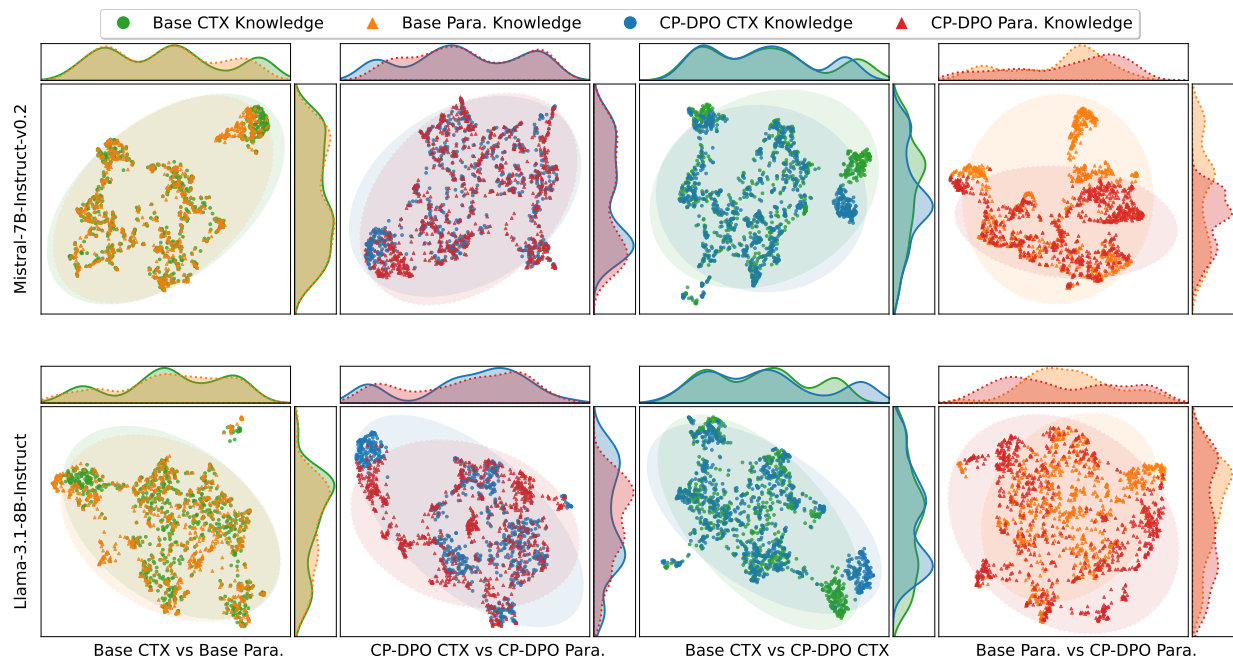


Fig. 9: Dimensionality reduction visualization of hidden states distributions between contextual (CTX) and parametric (Para.) knowledge on RAGTruth dataset across two base models. Each subplot shows pairwise comparisons with marginal KDE distributions and confidence ellipses.

CopyPaste-Link

Instruction: You are a professional text organization expert. Generate concise transition sentences to connect the core sentences and make the response flow naturally.

Query {*query*}

Core Sentences {*numbered_sentences*}

Requirements

1. Transition sentences should be concise (no more than 15 words)
2. They should logically connect adjacent core sentences
3. Focus on creating smooth flow between ideas
4. Common types: progression, contrast, addition, conclusion

Output Format

[TRANSITION_1_2]transition sentence
content[/TRANSITION_1_2]

[TRANSITION_2_3]transition sentence
content[/TRANSITION_2_3]

...

Optionally add:

[INTRO]introduction sentence[/INTRO]
[CONCLUSION]conclusion sentence[/CONCLUSION]

Please generate transitions:

Copying Requirements

1. RELEVANT CONTEXT REUSE: Incorporate relevant text.
2. MINIMAL ORIGINAL CONTENT: Limit additions to essential connections only.
3. PRESERVE EXACT WORDING: Keep original phrases and expressions.
4. CONTEXT-ONLY INFORMATION: Use only facts explicitly in the context, do not make up any information.
5. KEEP FLUENT and NATURAL ENGLISH.

Writer w/o Reviewer's Suggestions

Instruction: You are writer, skilled at copying relevant content from context to answer user questions. Generate highly copying responses from the given context.

Query

{*query*}

Context

{*context*}

Copying Requirements

{*copying_requirements*}

Answer:

Writer WITH Reviewer’s Suggestions

Instruction: You are Writer, skilled at copying relevant content from context to answer user questions. The Reviewer has suggested revisions to your old answer. Please provide a better answer to improve copying score and query relevance.

Your previous answer and Reviewer’s suggestions

Old Answer

{old_answer}

Reviewer’s Suggestions

{reviewer_suggestions}

Context

{context}

Query

{query}

Copying Requirements

{copying_requirements}

Answer:

Attributed

Instruction: Bear in mind that your answer should be strictly based on the following context.

Context: {context}

Query: {query}

Answer:

Citations

Instruction: Bear in mind that your answer should be strictly based on the following numbered passages. Add citations in square brackets [1], [2, 3], etc. at the end of sentences that are supported by the evidence.

Numbered Sentences

{numbered_sentences}

Query

{query}

Answer:

Reviewer

Your task is to review the answer to the query and suggest revisions with the goal of improving the answer’s copying score (contextual faithfulness) and query relevance.

Context

{context}

Query

{query}

Answer Awaiting Review

{answer}

Review Criteria

- Copying Score: Text reuse from context (Current: {copying_score})

- If copying score $\leq$ {copying_threshold}, require more context incorporation

- Contextual Faithfulness: All facts sourced from context only

- Remove any facts or knowledge not in context

- Reduce excessive or unnecessary original content

- Query Relevance: Direct addressing of user query

Provide CONCISE and ACTIONABLE suggestions (max 3 points):

C. Prompts of LLM Judges

We design the pairwise-comparison template and instructions to enable systematic, fine-grained evaluation of hallucinations in RAG responses.

Pairwise Comparison Template

Instruction: You are an expert judge. Compare two RAG responses (Response A and Response B)

{instruction}

Context: {context}

Response A: {response_a}

Response B: {response_b}

Please note: Do not question or doubt the provided context. Assume the context is absolutely correct, and make your verdict strictly based on this premise.

Output Format: {{ “verdict”: “<A/B/TIE>” }}

Above template is method-agnostic: it presents two anonymous responses, a common context treated as ground truth, and requires judges to output a formatted verdict—A, B or Tie.

The three instructions below can be slotted into the {instruction} placeholder in the above template and each then serves to pick the response exhibiting fewer RAG hallucinations along its respective dimension. Fabrication focuses on statements that are wholly unanchored in the provided context. Information-Distortion focuses on statements that misalign with the explicitly given context. False-Association focuses on claims that misweave separate pieces of context into an unsupported whole.

B. Baselines of Prompt-based

Base

{query}

Instruction for Comparing **Twist** Hallucination

for information distortion hallucination. The Core Definition of Information Twist: Altering key information in the Context (e.g., numbers, timelines, subjects, conclusions).

Which has fewer information distortion hallucinations?

Instruction for Comparing **Causal** Hallucination

for causal hallucination. The Core Definition of Causal: Forcibly linking unrelated content in the Context to form new conclusions unsupported by the Context.

Which has fewer false association hallucinations?

D. *Hit Rate and Accuracy*

Hit Rate

Context: $\{context\}$

Question: $\{question\}$

Based on the context, let's think step-by-step and answer the question in detail. Answer:

Accuracy

Context: $\{context\}$

Question: $\{question\}$

Options: $\{options\}$

Based on the above context, answer the question. You must output only a single token: A, B C or D. Do not provide any explanation or reasoning, just the chosen option. Answer: